

TAKlite Access Control Guide

Version: TAKlite v0.2.24

Purpose

This guide explains how TAKlite controls who can see Position Location Information (PLI), who can send datapackages, and who can receive datapackages.

The current access model is intentionally simple:

```
Team boundary first
Level second
Access override only when needed
Team links only when teams should interact
```

For most users, admins should assign only:

- Team
- Level

Most users do not need an access override.

Default Open Mode

TAKlite stays open until access policy is actually assigned.

That means:

- New server with no team assignments: users can see and send normally.
- Teams created but nobody assigned yet: users can still see and send normally.
- Once users are assigned to teams, team and level rules begin shaping traffic.

This keeps a simple TAKlite relay easy to use while still allowing controlled visibility when an admin needs it.

Core Terms

Connection User

A Connection User is a TAKlite user profile with an ATAK/WinTAK connection package.

Connection users are created in the `Users` panel and managed in the `Access` panel.

Team

A Team is the main visibility boundary.

Examples:

Alpha
Bravo
Charlie
Staff
Visitors

Users in the same team can see, send, and receive according to their level.

Users in different teams cannot see, send, or receive across those teams unless a team link or access override allows it.

Level

Levels are numeric access ranks from 1 to 4.

Levels apply only after the team path is allowed.

Level 1 sees/sends to Level 1
Level 2 sees/sends to Level 2 and Level 1
Level 3 sees/sends to Level 3, Level 2, and Level 1
Level 4 sees/sends to Level 4, Level 3, Level 2, and Level 1

Level does not jump team boundaries.

Example:

Alpha Level 4 cannot see Bravo Level 1
unless Alpha is linked to Bravo
or the Alpha user has an access override that allows all teams.

Team User

`Team User` is the default access type.

A Team User:

- Uses assigned teams as the boundary
- Uses level for rank filtering inside allowed teams
- Does not have global powers

This is the correct access type for normal users.

Access Override

An Access Override is an advanced template for high-trust users.

Use overrides for users who need capabilities beyond normal team and level rules.

Examples:

Controller
Instructor
Safety
Admin
Observer

Overrides can grant broad powers such as:

- See all teams
- Send to all teams
- Receive from all teams
- See assigned teams
- Send to assigned teams
- Receive from assigned teams

Important: an override controls what that user can do. It does not automatically make that user visible to everyone else.

The Most Important Rule

Team boundaries come first.

Levels only apply after the team path is allowed.

This means:

Team A Level 4 cannot see Team B Level 1 by level alone.

To make Team A interact with Team B, use a Team Link or an Access Override.

Level Example

Scenario:

- Controller users have an override for `See all teams` and `Send to all teams`.
- Controllers are Level 4.
- Team A users are Team Users.
- Team A users are Level 1.

Result:

- Controllers can see Team A users.
- Controllers can send to Team A users.
- Team A users cannot see Controllers unless their own team/link/override/level path allows it.
- Team A users cannot send to Controllers unless their own policy allows it.

This supports the common pattern:

High-trust users see everyone.
Normal users do not automatically see high-trust users.

Team Links

Team Links connect separate teams.

No link means separate teams stay separate.

Team links can be one-way or two-way.

No Link

Alpha Bravo

Alpha and Bravo stay isolated.

One-Way Link

Alpha -> Bravo

Alpha users can interact with Bravo users according to level.

Bravo users do not automatically interact with Alpha users.

Two-Way Link

Alpha <-> Bravo

Both teams can interact according to level.

Datapackage Behavior

TAKlite enforces the same access policy on datapackage search, query, download, and plugin-controlled send operations.

The backend policy is the failsafe:

- If a user should not see a package, it should not appear in their query/search results.
- If a user tries to download a blocked package, TAKlite should deny it.
- If a user sends through the Axon plugin, TAKlite checks the requested audience before delivery.

The Axon plugin can provide a pre-send review, but TAKlite still enforces policy server-side.

Recommended Admin Workflow

Use this order:

1. Create teams.
2. Create users.
3. Bulk assign users to teams.
4. Bulk set levels.
5. Leave most users as `Team User`.
6. Create access overrides only for controller/admin-style users.
7. Link teams only when cross-team interaction is needed.
8. Use Access Preview before field testing.

Creating Users

In `Users`:

1. Create a single user or bulk users.
2. Leave `Access type` as `Team User (default)` unless the user needs special powers.
3. Set the user's level, usually Level 1 for normal users.

4. Assign teams directly during creation or later in `Access`.

TAKlite learns IP/device details from TAK/Axon traffic when possible. Admins can edit those fields later if needed.

Bulk Membership

`Bulk Membership` is the primary admin workflow for many users.

Use it to:

- Select several users at once
- Replace/add/remove teams
- Set levels
- Set an access override
- Clear an access override back to Team User

Bulk Access Type Actions

Leave access type unchanged

Does not change whether selected users are Team Users or have an override.

Set to Team User

Clears any existing access override.

Set override

Applies the selected access override.

Bulk Team Actions

Replace teams

Removes existing team membership and applies the selected teams.

Add teams

Keeps existing teams and adds the selected teams.

Remove teams

Removes only the selected teams.

Individual Membership

Use `Individual Membership` when one user needs a manual correction.

You can edit:

- Access type
- Level
- Teams

Use this for cleanup, not primary onboarding.

Access Preview

Access Preview is the admin's sanity check.

Pick a user and verify:

- Who they can see
- Who they can send to
- Who can see them
- Who can send to them

Use Access Preview before issuing users their final connection packages.

Example 1: Two Isolated Teams

Goal:

- Alpha sees Alpha only.
- Bravo sees Bravo only.

Setup:

Teams:
Alpha
Bravo

Alpha users:
Access type: Team User

Team: Alpha
Level: 1

Bravo users:
Access type: Team User
Team: Bravo
Level: 1

Team links:
none

Result:

- Alpha users see Alpha users.
- Bravo users see Bravo users.
- Alpha and Bravo do not see each other.

Example 2: Controllers See Everyone

Goal:

- Controllers see all teams.
- Normal users do not see controllers.

Setup:

Access override:
Name: Controller
See all teams: On
Send to all teams: On
Receive from all teams: On

Controller users:
Access type: Controller
Level: 4
Team: optional

Normal users:
Access type: Team User
Level: 1
Team: Alpha or Bravo

Result:

- Controllers see Alpha and Bravo.
- Controllers can send to Alpha and Bravo.
- Alpha/Bravo users do not automatically see controllers.

Example 3: Temporary Cross-Team Visibility

Goal:

- Alpha and Bravo are normally isolated.
- For one event, Alpha and Bravo need to interact.

Setup:

Before event:

No link between Alpha and Bravo

During event:

Team Link: Alpha <-> Bravo

After event:

Set link back to No Link

Result:

- Teams can be merged or split without editing every user.
- Levels still apply across the linked path.

Example 4: One-Way Observation

Goal:

- Staff can see Student Team.
- Student Team cannot see Staff.

Setup:

Staff users:

Access type: Controller or Staff override

Level: 4

Student users:

Access type: Team User

Team: Student Team

Level: 1

Result:

- Staff sees students.
- Students do not see staff.

Alternate setup:

Teams:
Staff
Student Team

Team Link:
Staff -> Student Team

This keeps Staff as Team Users but still allows one-way team visibility.

Example 5: Mixed Levels Inside One Team

Goal:

- Team Lead sees everyone in Alpha.
- Alpha members do not all see the Team Lead.

Setup:

Team Lead:
Access type: Team User
Team: Alpha
Level: 4

Alpha normal users:
Access type: Team User
Team: Alpha
Level: 1

Result:

- Team Lead sees all Alpha users.
- Level 1 users see other Level 1 users.
- Level 1 users do not see the Level 4 Team Lead.

Common Mistakes

Mistake: Giving Everyone An Override

Most users should be Team Users.

Use overrides only for users who need broad power.

Mistake: Expecting Level To Cross Teams

Level does not cross teams by itself.

Use Team Links or an override for cross-team behavior.

Mistake: Forgetting Access Preview

Always preview one representative user from each team before field use.

Quick Checklist

Before handing out connection packages:

- Teams created
- Users assigned to correct teams
- Users assigned to correct levels
- Normal users left as Team User
- Controller/admin users given the intended override
- Team links set only where needed
- Access Preview checked for each team
- Test two real TAK clients before scaling up